

Rajendra Purohit

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SKILLS

Languages : Python, JavaScript ,SQL, HTML/CSS, MATLAB, Java
ML Libraries : Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn,PyTorch
AI/LLM : LangChain, LangGraph, RAG, Ollama, Prompt Engineering
Libraries/Frameworks : FastAPI, AngularJS
Tools : VS Code, Simulink, ControlDesk, ConfigurationDesk, MySQL

EXPERIENCE

IIT Jodhpur

Sep.2024 – Aug.2025

Junior Research Fellow (JRF)

Jodhpur,Rajasthan

- Developed and evaluated 15+ advanced machine learning and signal processing algorithms, heavily utilizing Backpropagation, Gradient Descent, Radial Basis Functions(RBF), Least Mean Square(LMS).
- Implemented sophisticated adaptive filtering models (including FxLMS, FeLMS, FxRLS, and Spline filters) to minimize the error functions, successfully improving overall **computational efficiency by 35%**.
- Converted complex mathematical models into functional logic for Active Noise Control (ANC) systems, establishing a strong foundation in the mathematical principles driving modern AI(Artificial Intelligence).
- Prototyped and deployed these optimized algorithms in real-time environments (using MATLAB/Simulink and dSPACE), achieving a highly accurate **10 dB noise reduction metric in physical test setups**.

MOBA Mobile Automation

Jan. 2024 – Apr. 2024

AngularJS Intern

Ahmedabad, Gujarat

- Developed responsive web interfaces using **AngularJS, JavaScript, HTML and CSS**.
- Improved UI performance and application speed through efficient DOM manipulation and implementation of reusable code components.

PROJECTS

Custom Image Classification CNN

- Architected and implemented a custom multi- layered Convolutional Neural Network (CNN) utilizing TensorFlow and Keras, incorporating advanced feature extraction through progressive convolutional filters and spatial dimensions reduction.
- Optimized model performance and mitigated overfitting by strategically integrating high-ratio dropout regularization, successfully achieving a robust 88.6% validation accuracy with exceptional generalization capabilities on image classification data.

Academic Performance Prediction

- Developed a multi-class Logistic Regression model to categorize student outcomes (high, medium, low), enabling data driven insights for potential academic intervention.
- Optimized model evaluation methodology, shifting from standard accuracy to precision-recall metrics to better reflect performance on real-world, imbalanced educational data.

Smart Mail Assistance

- Built a Python based email assistant that connects to inbox using IMAP to automatically fetch and parse recent mails.
- Using QyrouNnet model to generate automated, concise summaries of parsed email content.
- Designed and implemented a relational database schema (SQL) to store extracted email metadata alongside generated summaries for efficient querying and retrieval.

EDUCATION

Government Engineering College, Modasa

Aug. 2020 – Jun. 2024

Bachelor of Engineering in Information Technology

Modasa, Gujarat

- Coursework**: Data Structures, Algorithms, Databases, Operating Systems, Computer Networks
- CGPA**: 7.06